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BT 101 M.C.Q Midterm

BT 101

LECTURE 02

1-Water+light=?

(a) Biological energy (b) mechanical energy (c) **chemical energy** (d) none

2-Chemical energy+carbondioxide=?

(a) lipids (b) carbohydrates (c) **sugar** (d) none

LECTURE 03

1-How many types of the producers?

(A) 4 (b) 1 (c) **2** (d) 3

2-Synthesis of the organic compounds of their bodies from inorganic precursors such as:

- CO₂, H₂O, NO₃ known as the?

(a) Photoautotrophs (b) **autotrophs** (c) heterotrophs (d) chemoautotrophs

3-Examples of the photoautotrophs?

(a) plants & algae (b) algae & cyanobacteria (c) **cyanobacteria ,plants & algae** (d) fungi & cyanobacteria

4- The autotrophs which occurs in dark habitats & Convert toxic molecules into glucose known as the?

(a) Chemotrophs (b) prototroph (c) **Chemoautotrophs** (d) heterotrophs

Lecture no 04

1-Plant based is which type of the energy source for the herbivores?

(a) Secondary (b) **primary** (c) tertiary (d) none

2-Main types of the herbivores depending upon their energy source?

(a) 7 (b) 5 (c) **8** (d) 11

3-Main types of the carnivores?

- (a) **3** (b) 5 (c) 7 (d) 2

4-Primary decomposer?

- (a) Bacteria **(b) Fungi** (c) bacteria & fungi (d) none

5-Breaking down of tissues by the body's own internal chemicals and enzymes is called?

- (a) **Autolysis** (b) Heterolysis (c) Digestion (d) none

6-The process of the putrefaction performed by the bacteria release which compounds?

- (a) cadaverine and putrescine (b) putrescine (c) both **(d) a&c**

Lecture 05

1-Decomposition plays an important role in which cycles?

- (a) C&N** (b) C&O (c) C&N&O (d) N&P

2-The process of the proteolysis release the?

- (a) N&O **(b) N&P** (c) Both (d) None

Lecture No 06

1-Linear network of links to pass energy in form of food?

- (a) Trophic level (b) Feeding level (c) **Food chain** (d) none

2-The net effect of direct and indirect relations is called?

- (a) **Trophic cascades** (b) consumer-resource (c) producer-consumer (d) none

Lecture 08

1-Raw rate at which the primary producers synthesize new organic matter?

- (a) Primary productivity (PP) **(b) Gross primary productivity (GPP)**
(b) Net primary productivity (d) None

Lecture 09

1-The amount of the solar energy which is consumed in the photosynthesis?

- (a) 5% **(b) 1%** (c) 90% (d) 99%

2-The amount of the solar energy which is converted into the heat?

- (a) 97% (b) 90% **(c) 99%** (d) none

LECTURE NO 10

1-Flow of energy and decline of chemical-bond energy is represented by?

- (a) Photosynthetic plankton (b) phytoplankton **(c) pyramids** (d) none

LECTURE NO 11

1-The natural environment where species / organisms live?

- (a) Shelter (b) protection **(c) Habitat** (d) none

2-Orientation towards a source is called **(Taxis).**

LECTURE NO12

1-The condition in which the decreased body metabolism & lowered body temperature is known as the?

- (a) Winter sleep (b) Hibernation **(c) Tropor** (d) Aestivation

LECTURE NO 13

1- The water is essential factor in?

- (a) Evaporation (b) Excretion **(c) Homeostasis** (d) none

LECTURE NO 14

1-The house fly produce how many eggs through life time?

- (a) 500 eggs (b) 900 eggs **(c) 600 eggs** (d) none

2- Which is the type of the survivorship curve?

- (a) Convex & diagonal (b) Concave **(c) Both a&b** (d) none

LECTURE NO 15

1-How many types of the population growth?

- (a) **2** (b) 4 (c) 6 (d) 5

LECTURE NO 16

1-Which type of the population growth produce the S shaped curve?

- (a) **Logistic** (b)exponential (c) A&B (d) none

LECTURE NO 17

1-Which species has the high growth rate?

- (a)k-selected species (b) **r-selected species** (c) f-selected species (d) none

2-The examples of the k-selected species?

- (a)cephalopods & humans (b) rodents & parrots (c) **whales,humans& eagles** (d) A&B both

LECTURE NO 18

1-Oscillations are produced by interactions between populations of at least how many species?

- (a)3 (b) 5 (c) **2** (d) 4

LECTURE NO 20

1- The competition occurs between two species to use the same resources is called?

- (a) Intraspecific competition (b) **Interspecific competition** (c) both (d) none

LECTURE NO 21

1-The first community to become established in an area is called the?

- (a) Climax community (b) **Pioneer community** (c) Complex ecosystem (d) none

2-Each successional stage is called the **seralstage** .

3- The last community to be established in an area is called the?

- (a) **Climax community** (b) Pioneer community (c) Complex ecosystem (d) none

4- An event reduces already established ecosystem is called?

- (a) **Secondary succession** (b) Primary succession (c) Seral stage (d) None

LECTURE NO 22

1- Mutualistic gut flora are known as the

- (a) Cell wall digesting bacteria (b) **Cellulose digestingbacteria** (c) Glycogen digesting bacteria (d) none

2- Feed on non-grass plant material is known as the?

- (a) Coprography (b) over browsing (c) **browser** (d) gramnivory

LECTURE NO 23

1- How many stages form the foraging cycle?

(a) 5 (b) 3 (c) 4 (d) 6

2- Albatross make foraging flights within the prey range of?

(a) 700-300km (b) 3000-700km (c) 700-3000km (d) 300-700km

3-How many types of the inter& the intra specific competition?

(a) Direct (b) indirect (c) direct & indirect (d) none

LECTURE NO 25

1- Two or more species living in same habitat, reciprocally affect each other's evolution?

(A) Mutualism (c) co-evolution (c) commensalism (d) none

2- Gongylidia are the types of the ?

(a) Bacterial swellings (b) fungal swellings (c) both a& b (d) none

LECTURE NO 26

1-How many types of the symbiosis?

(a) 2 (b) 3 (c) 5 (d) none

2-Remora fish & sharks have the which type of the relationship?

(a) facultative (b) mutualism (c) commensalism (d) both a & c

3-How many types of the parasites?

(a) 2 (B) 5 (C) 4 (d) 3

LECTURE NO 27

1-Which strategy of the parasitism involves the intermediate host?

(a) Parasitic castrators (b) Directly transmitted (c) Vector transmitted (d) Parasitoids

LECTURE NO 28

1-How many types of the principles of the camouflage?

(a) 2 (b) 4 (c) 3 (d) 4

3- Use of any combination of materials, coloration, or illumination is known as the?

(a) cryptic coloration (b) camouflage (c) both a& b (d) none

LECTURE NO 29

1- The phenomena of the countershading which exhibits?

(a) Decorator crabs (c) grasshoppers (b) gazelles (d) both a& b

LECTURE NO 31

1-The resemblance of one species to another for protective or aggressive purposes is known as the?

(a) Camouflage (b) mimicry (c) aposematism (d) none

2-How many types of the mimicry?

(a) 4 (b) 2 (c) both (d) 3

3-Polymorphism is common in?

(a) Mullerian Mimicry (b) Batesian Mimicry (c) Model (d) none

LECTURE NO 32

1-Effective colors in the primary signals of the aposematism are?

red, yellow, black (b) yellow, black and white (c) white (d) all of the given

LECTURE NO 33

1-First atmosphere. It was aged about the?

(a) 1 million years ago (b) 3.8 million years ago (c) 3.8 billion years ago (d) 1 billion years ago

LECTURE NO 34

1-541 million years ago, O₂ made stable about which % of atmosphere?

(a) 10 % (b) 5% (c) 15% (d) 75%

LECTURE NO 35

1-Which are known as the permanent gases?

(a) 78% Nitrogen (N₂) • 21% Oxygen (O₂) (b) Carbon Dioxide (CO₂): 0.038% • Methane (CH₄): 0.00017%

(b) <1% Argon (Ar) (d) both a & c

2-Relative percentages of the permanent gases remain constant up to?

(a) 80-100m (b) 100-80km (c) 80-100km (d) none

3-Turbulent mixing causes atmospheric composition to be fairly homogeneous. Called as?

(a) Hemisphere (b) atmosphere (c) homosphere (d) none

4-Typical mid latitude value is?

(A)4% (b) 2% **(c) 1-2%** (d) 2-1%

5-Aerosols can be?

(a) Manmade (b) naturally occurring **(c) both of these** (d) none

LECTURE NO 36

1-Atmosphere can be divided into the main?

(a) 3 layers (b) 2 layers **(c) 5 layers** (d) 7 layers

LECTURE NO 37

1-The total average height of the troposphere is?

(a)18 km (b) 6km (c) 17km **(d) 13km**

2-Temperature of the troposphere with altitude remains?

(a) Constant **(b) decreases** (c) increases (d) none

LECTURE NO 38

1-Ozone is important gas in which part of the atmosphere?

(a)mesosphere (b) troposphere **(c) stratosphere** (d) thermosphere

2- Bacterial life survives in which sphere making it a part of the biosphere?

(a) Troposphere **(b) stratosphere** (c) mesosphere (d) none

3-The temperature in the stratosphere ranges from the?

(a)-51 degree (b) 15 degree **(c) – 15 Celsius** (d) both B&C

LECTURE NO 39

1-Which part of the atmosphere is considered as the "near space"

(a) Thermosphere (b) stratosphere **(c) mesosphere** (d) troposphere

2-The process of the ionization is observed in which part of the atmosphere?

(a) Thermosphere (b) stratosphere **(c) mesosphere** (d) troposphere

LECTURE NO 40

1-The thermosphere & the exosphere are known as the?

- (a) "near space" (b) ionosphere (c) both (d) none

LECTURE NO 41

1- CO₂, methane, nitrous oxide cause?

- (a) Global warming (b) greenhouse effect (c) both (d) none

LECTURE NO 42

1- Human body is carbon by weight.

- (a) 5% (b) 15% (c) 20% (d) 3%

2- Dissolved CO₂ and bicarbonates are used by algae and aquatic plants in the process of the?

- (a) Respiration (b) photosynthesis (c) both (d) none

LECTURE NO 43

1- When a gas is changed into a liquid this process is called?

- (a) precipitation (b) evaporation (c) condensation (d) A&C both

2-Adult human body contains how many % of water by weight?

- (a) 90% (b) 65% (c) 60% (d) 75%

3-The process of the condensation is opposite of the process of the?

- (a) Precipitation (b) surface runoff (c) evaporation (d) none

LECTURE NO 45

1-A living organism contain nitrogen by weight?

- (a) 18% (b) 75% (c) 16% (d) both b&c

2-How many phases of the nitrogen cycle?

- (a) 6 (b) 5 (c) 9 (d) 10

LECTURE NO 46

1-C/N typical for soil microbial biomass is?

- (a) 35 (b) 20 (c) 25 (d) none

2-In the process of the nitrification the 2nd step is catalyzed by the?

(a) Nitosomnas (b) **Nitrobacter** (c) Rhizobium legume (d) none

LECTURE NO 48

1- How many % of the extinction species of the animals?

(a) 20 (b) 35 (c) **19** (d) 5

2-Natural habitats may be adversely affected by humans in?

(a) 5 (b) **3** (c) 7 (d) 4

LECTURE NO 49

1-How Earth is covered by the forests?

(a) 50% (b) **31%** (c) 90% (d) none

2-Tropical deforestation is responsible for approximately % of world greenhouse gas emissions?

(a) 25% (b) **20 %** (c) 31% (d) 90%

LECTURE NO 52

1-Lionfish introduced into the united states through the aquarium trade in?

(a) 1985 (b) 1800 (c) 1998 (d) **1992**

LECTURE NO 54

1-Pollution can be classified as the how many points?

(a) 4 (b) **2** (c) 3 (d) 5

LECTURE NO 56

1-In water pollution the level of the BOD increases the level of the DO remains the?

(a) Increase (b) **decreases** (c) remains constant (d) none

LECTURE NO 57

1-Mercury bioaccumulates in the muscles of the top predators of the?

(a) Closed ocean (b) **open ocean** (c) both (d) none

2-The substances which Contain atoms of unstable isotopes that spontaneously emit radiation are known as the?

- (a) Nuclear substances (b) organic substances **(c) radioactive substances** (d) inorganic substances

LECTURE NO 58

1-How many major classes of the air pollution?

- (a) 2 **(b) 6** (c) 5 (d) 7

LECTURE NO 59

1-Diverse group of organic compounds that contain only hydrogen and carbon (ex: CH₄-methane):

- (a) Carbon oxides **(b) hydrocarbons** (c) both (d) none

2-Man-made pollutant in the lower atmosphere is known as the?

- (a) Ozone **(b) Tropospheric Ozone** (c) Stratospheric Ozone (d) none

LECTURE NO 60

1-Which are strongly affected by the light pollution?

- (a) Amphibians (c) Birds **(c) reptiles** (d) animals

LECTURE NO 62

1-Unpolluted rain has an acidic pH, but usually no lower than?

- (a) 5 (b) 3 **(c) 5.7** (d) 7

2-Volcanic eruption releases in air?

- (a) NO₂ **(b) SO₂** (c) SO₃ (d) NO₃

LECTURE NO 63

1-How many stages of the water cycle?

- (a) 3 **(b) 6** (c) 7 (d) 9

2-Deforestation directly affects the rate of the?

- (a) Condensation (b) evaporation (c) precipitation **(d) transpiration**

3-How many people live in the coastal areas?

- (a) 5 billion **(b) 3 billion** (c) 7 billion (d) 1 billion

4-Depletion of oxygen in water disturbing the?

(a) Algae (b) fauna (c) flora **(d) fauna & flora**

LECTURE NO 64

1-How many % of the greenhouse effect in the World?

(a) 25% (b) 40% **(c) 36-70%** (d) 3.5 %

2-Which of them act as the greenhouse gases?

(a) CO₂ (b) methane (c) methane, nitrous oxide & water vapors **(d) both A & C**

3-Methane is how many times effective than that of the CO₂?

(a) 15 (b) 5 **(c) 20** (d) 25

NO 65 LECTURE

1-Ozone absorb a band of the ultraviolet radiation is called?

(a) UV (b) UB **(c) UVB** (d) none

2-Gas is composed of how many atoms of the oxygen?

(a) 5 **(b) 3** (c) 1 (d) 4

3-Cyanobacteria are sensitive to the indirectly affecting the crops?

(a) UVA (b) UVB **(c) UV** (d) none

LECTURE NO 68

1-Oxygen is less soluble in the?

(a) **Warm water** (b) cold water (c) both (d) none

LECTURE NO 69

1-Increasing concentration of toxic substance?

(a) Bioaccumulation **(b) bio magnification** (c) both (d) none

2-Biomagnification occurs?

(a) Within trophic level **(b) across trophic level** (c) both (d) none

3-Pollutants that undergo the bio magnification?

(a) DDT (b) HCB (c) PCBs **(d) all of the given**

4-DDT interferes which metabolism?

(a) Magnesium (b) iron **(c) Calcium** (d) none

5-In which country DDT is used to overseas to prevent the malaria?

(a)US **(b) Africa** (c) Pakistan (d) both a&b

LECTURE NO 70

1-How many types of the waste water according to the usage?

(a) **5** (b) 3 (c) 9 (d)7

2-Global fresh water use in agriculture?

(a)35% (b)69% (c) 90% **(d) both B&C**

3-How many types of the water on the basis of the contamination?

(a)3 **(b)2** (c) 5 (d) 1

LECTURE NO 72

1-How deep is lagoons (stabilization ponds)?

(a)5-3' **(b) 3-5'** (c) both (d) none

2-How many stages of the conventional sewage treatment?

(a)3 (b) 5 (c)1 (d) 2

LECTURE NO 73

1-Activated sludge-sewage mixture is called? **(MIXED LIQUOR)**

LECTURE NO 74

1-Pere David's deer was a native species of?

(a) UK (b) England **(c) China** (c) Africa

LECTURE NO 75

1-First evolutionary scheme was proposed by Jean-Baptiste Lamarck in?

(a)1735 **(b) 1809** (c)1880 (d) 1709

2-The theory of the natural selection proposed by the Darwin in?

(a)1798 (b) 1865 **(c)1859** (d)1809

3-Gregor Mendel proposed the law of the genetics in?

(A) 1798 **(b) 1865** (c) 1859 (d) 1809

LECTURE NO 76

1-Regions with identical climate have different floras and faunas?

(a) Darwin's law **(b) Buffon law** (c) Mendel law (d) none

2-Barier is classified in how man terms?

(a) 4 **(b) 3** (c) 5 (d) 2

LECTURE NO 77

1-Nicolas Steno (1638-86) Father of?

(a) Evolution & Paleontology **(b) geology and stratigraphy** (c) both a & b (d) none

LECTURE NO 78

1-Comparative anatomy is the study of similarities and differences in the anatomy of?

(a) Same species **(b) different species** (c) both (d) none

LECTURE NO 79

1-Genetic code of mRNA **is universal**

LECTURE NO 80

1- If an organ is used a lot it will develop and strengthen. •If it is not used it will atrophy is called the law of?

(a) Law of the use (b) law of the disuse **(c) both** (d) none

2-Which changes only transformed to the next generation?

(a) Germ cells (b) DNA (c) RNA **(d) Both a & b**